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Reversible electromagnetic valve, and air conditioning unit having same

Abstract

A reversible electromagnetic valve and an air conditioning unit are provided. The reversible electromagnetic valve includes a valve body and a valve core assembly. A first communication port and a second communication port are respectively formed in two sides of the valve body provided with a valve cavity. The valve core assembly is capable of slidably moving in the valve cavity and includes a connecting rod, a slider unit and a piston unit. The slider unit and the piston unit are respectively connected to two ends of the connecting rod. The slider unit is arranged adjacent to the first communication port and the second communication port opposite to each other. The slider unit is configured to block both the first communication port and the second communication port and the piston unit is capable to be pushed to slide in the valve cavity.

Inventors: Feng; Zhongbo (Shaoxing, CN), Xiong; Yunjun (Shaoxing, CN)

Applicant: ZHEJIANG DUNAN ARTIFICIAL ENVIRONMENT CO., LTD. (Shaoxing, CN)

Family ID: 84141129

Assignee: ZHEJIANG DUNAN ARTIFICIAL ENVIRONMENT CO., LTD. (Shaoxing, CN)

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Primary Examiner: Colon-Morales; David

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of international patent application No. PCT/CN2022/094038, filed on May 20, 2022, which itself claims priority to Chinese patent application Nos. 202121097730.X, filed on May 20, 2021, titled “REVERSIBLE ELECTROMAGNETIC VALVE, AND AIR CONDITIONING UNIT HAVING

SAME”, and 202111489240.9, filed on Dec. 7, 2021, titled “REVERSIBLE ELECTROMAGNETIC VALVE”, which are hereby incorporated by reference.

TECHNICAL FIELD

(1) The present disclosure relates to a field of refrigeration, and in particular, to a reversible electromagnetic valve and an air conditioning unit having the same.

BACKGROUND

(2) A reversible electromagnetic valve is usually installed in an air conditioning unit and used for achieving bidirectional circulation of a medium.

(3) A valve core assembly of a traditional reversible electromagnetic valve includes a connecting rod, a slider and two piston units. The two piston units are connected to two ends of the connecting rod respectively. In the process that the medium pushes the valve core assembly to move, the two piston units need to be pushed to slide, however, it has a large friction force when the piston units slide in a sealed valve cavity, a greater thrust is required for the medium pushing the two piston units to slide, therefore it needs a large valve opening pressure difference for the reversible electromagnetic valve in the related art.

SUMMARY

(4) According to various embodiments of the present disclosure, a reversible electromagnetic valve is provided.

(5) A reversible electromagnetic valve is provided, including a valve body and a valve core assembly. A first communication port and a second communication port are respectively formed in two sides of the valve body. The valve body is provided with a valve cavity. The valve core assembly is arranged in the valve cavity. The valve core assembly is capable of slidably moving in the valve cavity to enable the first communication port and the second communication port to be in communication with each other or out of communication with each other. The valve core assembly includes a piston unit, a connecting rod and a slider unit. The slider unit and the piston unit are respectively connected to two ends of the connecting rod. The slider unit is arranged adjacent to the first communication port and the second communication port. The first communication port and the second communication port are opposite to each other. The slider unit is configured to block the first communication port and the second communication port at the same time. The piston unit is capable to be pushed to slide in the valve cavity due to a pressure difference between two sides of the piston unit.

(6) In some embodiments, the reversible electromagnetic valve further includes an intermediate end cover and a second end cover. The intermediate end cover is fixed in the valve cavity. The piston unit is arranged on one side of the intermediate end cover away from the first communication port and the second communication port. A first cavity is defined by the intermediate end cover and the piston unit. A second cavity is defined by the piston unit and the second end cover.

(7) In some embodiments, the reversible electromagnetic valve further includes a pilot valve, disposed on the valve body. The valve body is provided with a first hole and a second hole. The first hole is in communication with the first cavity. The second hole is in communication with the second cavity. The pilot valve is respectively in communication with the first hole and the second hole through a capillary tube. The pilot valve is configured to control a pressure difference between the first cavity and the second cavity.

(8) In some embodiments, the reversible electromagnetic valve further includes a first end cover disposed at an end of the valve body away from the second end cover. A medium cavity is defined between the first end cover and the intermediate end cover. The slider unit is disposed in the medium cavity, and when the reversible electromagnetic valve is in a first position, the first communication port and the second communication port are respectively in communication with the medium cavity.

(9) In some embodiments, a side surface of the piston unit facing the intermediate end cover is

provided with a first protrusion, and when the piston unit moves towards the first communication port and the second communication port, the first protrusion is capable of abutting against the intermediate end cover; and/or a side surface of the piston unit facing the second end cover is provided with a second protrusion, and when the piston unit moves towards the second end cover, the second protrusion is capable of abutting against the second end cover.

(10) In some embodiments, a side surface of the piston unit facing the intermediate end cover is provided with a first protrusion, the first protrusion is in an annular shape, a side surface of the first protrusion is provided with a first throttling hole for allowing a medium to flow out of a groove formed by the first protrusion; and/or a side surface of the piston unit facing the second end cover is provided with a second protrusion, the second protrusion is in an annular shape, a side surface of the second protrusion is provided with a second throttling hole for allowing a medium to flow out of a groove formed by the second protrusion.

(11) In some embodiments, a side surface of the piston unit facing the intermediate end cover is provided with a first protrusion, the first protrusion is in an annular shape, a side surface of the first protrusion is provided with a first throttling groove for allowing a medium to flow out of a groove formed by the first protrusion; and/or a side surface of the piston unit facing the second end cover is provided with a second protrusion, the second protrusion is in an annular shape, a side surface of the second protrusion is provided with a second throttling groove for allowing a medium to flow out of a groove formed by the second protrusion.

(12) In some embodiments, the slider unit includes a first portion and a second portion opposite to each other, the first portion is capable of blocking the first communication port, the second portion is capable of blocking the second communication port, an accommodating cavity is formed between the first portion and the second portion, an elastic member is arranged in the accommodating cavity, and two ends of the elastic member abut against the first portion and the second portion respectively.

(13) In some embodiments, the valve cavity includes a medium cavity. The slider unit is arranged in the medium cavity. A balance hole is formed in the first portion and/or the second portion. The medium cavity is in communication with the accommodating cavity through the balance hole.

(14) In some embodiments, the slider unit further includes a guide frame. The guide frame is sleeved on an outer side of the first portion and the second portion and connected to the connecting rod.

(15) The present disclosure further provides an air conditioning unit including the above reversible electromagnetic valve.

(16) The details of one or more embodiments of the present disclosure are set forth in the following drawings and description. Other features, objects, and advantages of the present disclosure will become apparent from the description, drawings, and claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) To better describe and illustrate embodiments and/or examples of the present disclosure, reference may be made to one or more drawings. Additional details or examples for describing the figures should not be considered limiting of the scope of any of the present disclosure, presently described embodiments and/or examples, and the best modes of these applications presently being understood.

(2) FIG. 1 is a schematic view of a reversible electromagnetic valve in a related art.

(3) FIG. 2 is an overall structural diagram of a reversible electromagnetic valve in some embodiments of the present disclosure.

(4) FIG. 3 is a schematic view of a reversible electromagnetic valve in some embodiments of the

present disclosure.

(5) FIG. 4 is a schematic view of a throttling hole of a reversible electromagnetic valve in some embodiments of the present disclosure.

(6) FIG. 5 is a schematic cross-sectional view of a reversible electromagnetic valve in some embodiments of the present disclosure.

(7) FIG. 6 is a schematic view of a throttling groove of a reversible electromagnetic valve in some embodiments of the present disclosure.

(8) FIG. 7 is a schematic cross-sectional view of a reversible electromagnetic valve in some embodiments of the present disclosure.

(9) FIG. 8 is a schematic cross-sectional view of a reversible electromagnetic valve in some embodiments of the present disclosure.

(10) FIG. 9 is a partial enlarged schematic view of the reversible electromagnetic valve in FIG. 8.

(11) FIG. 10 is a schematic cross-sectional view of a first sealing seat of a reversible electromagnetic valve in some embodiments of the present disclosure.

(12) FIG. 11 is a schematic cross-sectional view of a second sealing seat of a reversible electromagnetic valve in some embodiments of the present disclosure.

(13) FIG. 12 is a schematic cross-sectional view of an intermediate end cover of a reversible electromagnetic valve in some embodiments of the present disclosure.

(14) FIG. 13 is a schematic diagram of a relationship between an air conditioning unit and a reversible electromagnetic valve in some embodiments of the present disclosure.

REFERENCE SIGNS

(15) **100** represents a reversible electromagnetic valve; **110** represents a pilot valve; **200** represents an air conditioning unit; **10** represents a valve body; **11** represents a first communication port; **111** represents a first connecting pipe; **12** represents a second communication port; **121** represents a second connecting pipe; **13** represents a valve cavity; **131** represents a first cavity; **132** represents a second cavity; **133** represents a medium cavity; **14** represents a first hole; **141** represents a first capillary tube; **15** represents a second hole; **151** represents a second capillary tube; **16** represents a valve base; **20** represents a valve core assembly; **21** represents a piston unit; **211** represents a first protrusion; **2111** represents a first throttling hole; **2112** represents a first throttling groove; **212** represents a second protrusion; **2121** represents a second throttling hole; **2122** represents a second throttling groove; **213** represents a first piston bowl; **214** represents a second piston bowl; **215** represents an intermediate baffle; **216** represents a first baffle; **217** represents a second baffle; **22** represents a connecting rod; **23** represents a slider unit; **231** represents a first portion; **2311** represents a balance hole; **232** represents a second portion; **233** represents an accommodating cavity; **234** represents an elastic member; **235** represents a guide frame; **30** represents an intermediate end cover; **31** represents a through hole; **311** represents a mounting groove; **312** represents a sealing ring; **32** represents a first end cover; **33** represents a second end cover; **40** represents a sleeving pipe; **50** represents a first sealing seat; **51** represents a first mounting groove; **511** represents an annular flange; **512** represents a third sealing groove; **52** represents a first through hole; **53** represents a first sealing groove; **54** represents a welding groove; **60** represents a second sealing seat; **61** represents a second through hole; **611** represents a second sealing groove; **70** represents a first sealing ring; **80** represents a second sealing ring; **90** represents a third sealing ring; **101** represents a reversible electromagnetic valve in the related art; **1011** represents a valve core assembly in the related art; **1012** represents a connecting rod in the related art; **1013** represents a piston unit in the related art; **1014** represents a slider in the related art; and **1015** represents a valve cavity in the related art.

DETAILED DESCRIPTION OF THE EMBODIMENT

(16) The technical solutions in the embodiments of this application will be clearly and completely described below with reference to the accompanying drawings in the embodiments of the present disclosure. All other embodiments obtained by a person of ordinary skill in the art based on the

embodiments of the present disclosure without creative efforts shall fall within the protection scope of the present disclosure.

(17) It should be noted that when the component is referred to as being “mounted on” another component, it may be directly on the other component or there may be a centered component. When one component is considered to be “disposed on” another component, it may be directly disposed on another component or may have a centering component simultaneously. When one component is considered to be “fixed to” another component, it may be directly fixed on another component or there may be a centering component at the same time.

(18) Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by one skilled in the art to which this application belongs. The terminology used herein in the specification of the present disclosure is for the purpose of describing particular embodiments only and is not intended to be limiting of the present disclosure. As used herein, the term “or/and” includes any and all combinations of one or more of the associated listed items.

(19) Referring to FIG. 13, an embodiment of the present disclosure provides a reversible electromagnetic valve **100**, which can be applied in an air conditioning unit **200**. The reversible electromagnetic valve **100** is configured to control a communication of the pipeline or cut off the communication of the pipeline, thereby achieving bidirectional circulation of a medium.

(20) The reversible electromagnetic valve **100** in the present disclosure can be used for an air conditioning unit **200** requiring bidirectional operation, for example, the reversible electromagnetic valve **100** can be mounted in the air conditioning unit **200** requiring two functions of a refrigeration mode and a heating mode, so as to realize bidirectional flow of a medium without switching a pipeline. The medium in the present disclosure can refer to a refrigerant.

(21) Referring to FIG. 1, a valve core assembly **1011** of a reversible electromagnetic valve **101** in the related art includes a connecting rod **1012**, a slider **1014** and two piston units **1013**. The two piston units **1013** are respectively connected to two ends of the connecting rod **1012**. In a process of a medium pushing the valve core assembly **1011**, the two piston units **1013** need to be pushed to slide, however, it has a large friction force when the piston units **1013** slide in a sealed valve cavity **1015**, a greater thrust is required for the medium pushing the two piston units **1013** to slide, therefore it needs a large valve opening pressure difference for the reversible electromagnetic valve in the related art.

Embodiment 1

(22) Referring to FIG. 3, the reversible electromagnetic valve **100** in the present disclosure includes a valve body **10** and a valve core assembly **20**. Two sides of the valve body **10** are respectively provided with a first communication port **11** and a second communication port **12**. The valve body **10** is provided with a valve cavity **13**. The valve core assembly **20** is arranged in the valve cavity **13**, and the valve core assembly **20** is capable of slidably moving in the valve cavity **13** to enable the first communication port **11** and the second communication port **12** to be in communication with each other or out of communication with each other.

(23) Further, a first connecting pipe **111** is provided in the first communication port **11**. A second connecting pipe **121** is provided in the second communication port **12**. And the first connecting pipe **111** and the second connecting pipe **121** are respectively connected to pipelines of the air conditioning unit.

(24) The reversible electromagnetic valve **100** further includes a valve base **16**. The valve base **16** is mounted in the valve cavity **13**. The first communication port **11** and the second communication port **12** penetrate the valve body **10** and the valve base **16**. And one end of the valve core assembly **20** is capable of moving in the valve base **16** to enable the first communication port **11** and the second communication port **12** to be in communication with each other or out of communication with each other.

(25) Referring to FIG. 3, the valve core assembly **20** includes a piston unit **21**, a connecting rod **22**

and a slider unit **23**. The slider unit **23** and the piston unit **21** are respectively connected to two ends of the connecting rod **22**. The slider unit **23** is disposed adjacent to the first communication port **11** and the second communication port **12**. And a pressure difference can be formed between two sides of the piston unit **21** to push the piston unit **21** to slide in the valve cavity **13**.

(26) By providing the single piston unit **21** to drive the slider unit **23** to move in the valve cavity **13**, the first communication port **11** and the second communication port **12** can be in communication with each other or out of communication with each other, and the thrust required for pushing the piston unit **21** can be reduced, so that a valve opening pressure difference required when the reversible electromagnetic valve **100** is opened can be reduced. The reliability of opening and closing the reversible electromagnetic valve **100** at a low pressure can be improved. Compared to two piston units of the reversible electromagnetic valve in the related art, a single piston unit **21** in the present disclosure is provided to reduce the number of components, so that the overall volume of the reversible electromagnetic valve **100** is reduced, the reversible electromagnetic valve **100** is miniaturized, materials are saved, and the processing cost is reduced.

(27) The first communication port **11** and the second communication port **12** are opposite to each other. The slider unit **23** is configured to block both the first communication port **11** and the second communication port **12**. When the reversible electromagnetic valve **100** is in a closed state, when the first communication port **11** is an inlet, the first communication port **11** is blocked by the slider unit **23** and cannot be in communication with the second communication port **12**. When the second communication port **12** is an inlet, the second communication port **12** is blocked by the slider unit **23** and cannot be in communication with the first communication port **11**, so that the reversible electromagnetic valve **100** can work effectively.

(28) In the present disclosure, the first communication port **11** of the reversible electromagnetic valve **100** can be an inlet, and after a mode of the air conditioning unit **200** is switched, the second communication port **12** can be an inlet, and when the reversible electromagnetic valve **100** is in a closed state, the slider unit **23** can block the first communication port **11** and the second communication port **12** at the same time, thereby preventing the medium from flushing the slider unit **23** and affecting a sealing performance.

(29) Furthermore, the reversible electromagnetic valve **100** includes an intermediate end cover **30** and a second end cover **33**. The intermediate end cover **30** is fixed in the valve cavity **13**. The piston unit **21** is disposed on a side of the intermediate end cover **30** away from the first communication port **11** and the second communication port **12**. A first cavity **131** is formed between the intermediate end cover **30** and the piston unit **21**. A second cavity **132** is formed between the piston unit **21** and the second end cover **33**. The first cavity **131** and the second cavity **132** are respectively located on two sides of the piston unit **21**. When a pressure difference between the first cavity **131** and the second cavity **132** is formed, the piston unit **21** can be pushed to move, so as to drive the slider unit **23** to enable the first communication port **11** and the second communication port **12** to be in communication with each other or out of communication with each other.

(30) Furthermore, the intermediate end cover **30** is provided with a through hole **31**. A mounting groove **311** is located on an inner wall of the through hole **31** and configured for allowing the sealing ring **312** to be mounted. The connecting rod **22** extends into the through hole **31**. A part of the connecting rod **22** extends towards the first communication port **11** and the second communication port **12**, and the other part of the connecting rod **22** extends towards the second end cover **33**. The connecting rod **22** is capable of moving in the through hole **31**. And the connecting rod **22** and the intermediate end cover **30** are sealed with each other through a sealing ring **312**.

(31) The reversible electromagnetic valve **100** further includes a first end cover **32**. The first end cover **32** is disposed at an end of the valve body **10** away from the second end cover **33**. A medium cavity **133** is formed between the first end cover **32** and the intermediate end cover **30**. The slider unit **23** is disposed in the medium cavity **133**. When the reversible electromagnetic valve **100** is in a

first position, and the first communication port **11** and the second communication port **12** are respectively in communication with the medium cavity **133**. It should be noted that the first position refers to a position of each component when the reversible electromagnetic valve **100** is in an opened state. The medium cavity **133** is configured for allowing the medium to flow. When the valve core assembly **20** slides away from the first communication port **11** and the second communication port **12**, the first communication port **11** can be in communication with the second communication port **12** through the medium cavity **133**, so that the medium achieves bidirectional circulation.

(32) Referring to FIG. 2 and FIG. 3, specifically, the first communication port **11** and the second communication port **12** are respectively located in the medium cavity **133**. The valve body **10** is provided with a first hole **14** and a second hole **15**. The first hole **14** is in communication with the first cavity **131**. The second hole **15** is in communication with the second cavity **132**. The first hole **14** is internally provided with a first capillary tube **141**. The second hole **15** is internally provided with a second capillary tube **151**. The first capillary tube **141** and the second capillary tube **151** are connected to a pilot valve **110**, so that a pressure difference between the first cavity **131** and the second cavity **132** is formed to push the piston unit **21** to move.

(33) When the reversible electromagnetic valve **100** needs to be closed, a high-pressure refrigerant enters the second cavity **132** from the second capillary tube **151**, and the refrigerant in the first cavity **131** flows out of the first capillary tube **141**, thereby pushing the piston unit **21** to move towards the first communication port **11** and the second communication port **12**. When the reversible electromagnetic valve **100** needs to be opened, the high-pressure refrigerant enters the first cavity **131** from the first capillary tube **141**, and the refrigerant in the second cavity **132** flows out of the second capillary tube **151**, thereby pushing the piston unit **21** to move towards the first communication port **11** and the second communication port **12**.

(34) Referring to FIG. 3, a side surface of the piston unit **21** facing the intermediate end cover **30** is provided with a first protrusion **211**. When the piston unit **21** moves towards the first communication port **11** and the second communication port **12**, the first protrusion **211** can abut against the intermediate end cover **30**, thereby reducing an impact force of the piston unit **21** on the intermediate end cover **30**.

(35) Referring to FIG. 4, furthermore, the first protrusion **211** is in an annular shape. A side wall of the first protrusion **211** is provided with a first throttling hole **2111**. When the piston unit **21** is pushed to move toward the first communication port **11** and the second communication port **12**, a part of the medium can slowly flow out of a groove formed by the first protrusion **211** through the first throttling hole **2111**, resulting in forming a resistance, so that an impact force of the piston unit **21** on the intermediate end cover **30** is slowed down, a mounting firmness of the intermediate end cover **30** is enhanced, and noise can be reduced.

(36) Referring to FIG. 5 and FIG. 6, in another embodiment, an end surface of the first protrusion **211** facing the intermediate end cover **30** is provided with a first throttling groove **2112**. When the piston unit **21** is pushed to move towards the first communication port **11** and the second communication port **12**, a part of the medium can slowly flow out of a groove formed by the first protrusion **211** through the first throttling groove **2112**, resulting in forming resistance. So that an impact force of the piston unit **21** on the intermediate end cover **30** is slowed down, an installation firmness of the intermediate end cover **30** is enhanced, and noise can be reduced.

(37) Referring to FIG. 3, in another embodiment, a side surface of the piston unit **21** facing the second end cover **33** is provided with a second protrusion **212**. When the piston unit **21** moves towards the second end cover **33**, the second protrusion **212** can abut against the second end cover **33**, thereby reducing an impact force of the piston unit **21** on the second end cover **33**.

(38) Furthermore, the second protrusion **212** is in an annular shape. A side surface of the second protrusion **212** is provided with a second throttling hole **2121**. When the piston unit **21** is pushed to move towards the second end cover **33**, a part of the medium can slowly flow out of a groove

formed by the second protrusion **212** through the second throttling hole **2121**, resulting in forming resistance. So that an impact force of the piston unit **21** on the second end cover **33** is reduced, and noise is reduced.

(39) Referring to FIG. 5, in another embodiment, an end surface of the second protrusion **212** facing the second end cover **33** is provided with a second throttling groove **2122**. When the piston unit **21** is pushed to move towards the second end cover **33**, a part of the medium can slowly flow out of a groove formed by the second protrusion **212** through the second throttling groove **2122**, resulting in forming resistance. So that an impact force of the piston unit **21** on the second end cover **33** is reduced, and noise is reduced.

(40) Specifically, the piston unit **21** includes a first piston bowl **213**, a second piston bowl **214**, an intermediate baffle **215**, a first baffle **216** and a second baffle **217**. The first baffle **216**, the intermediate baffle **215** and the second baffle **217** are connected to each other. The first baffle **216** and the second baffle **217** are located on two sides of the intermediate baffle **215**. The first baffle **216** is arranged adjacent to the intermediate end cover **30**. The first piston bowl **213** is located between the first baffle **216** and the intermediate baffle **215**. The second piston bowl **214** is located between the intermediate baffle **215** and the second baffle **217**. The first protrusion **211** is disposed on the first baffle **216**. And the second protrusion **212** is disposed on the second baffle **217**.

(41) Referring to FIG. 3, the slider unit **23** is disposed in the medium cavity **133**. The slider unit **23** includes a first portion **231** and a second portion **232** opposite to each other. The first portion **231** is capable of blocking the first communication port **11**. The second portion **232** is capable of blocking the second communication port **12**. An accommodating cavity **233** is provided between the first portion **231** and the second portion **232**. An elastic member **234** is disposed in the accommodating cavity **233**. Two ends of the elastic member **234** abut against the first portion **231** and the second portion **232** respectively. An interaction force between the elastic member **234** and the first portion **231** and an interaction force between the elastic member **234** and the second portion **232** can enable the first portion **231** and the second portion **232** to be separated from each other, so that the first portion **231** can block the first communication port **11** and the second portion **232** can block the second communication port **12**, a sealing performance during plugging can be improved.

(42) Furthermore, the first portion **231** and/or the second portion **232** are provided with a balance hole **2311**. The medium cavity **133** is in communication with the accommodating cavity **233** through the balance hole **2311**. And the balance hole **2311** can balance a pressure between the medium cavity **133** and the accommodating cavity **233** to prevent the medium from extruding the slider unit **23**, thereby ensuring stability of the slider unit **23**.

(43) Referring to FIG. 3, the slider unit **23** further includes a guide frame **235**. The guide frame **235** is sleeved on outer sides of the first portion **231** and the second portion **232** and connected to the connecting rod **22**. The guide frame **235** can play a role in limiting and guiding the slider unit **23**, so that the slider unit **23** is prevented from inclining under the action of the medium, and sealing performance between the slider unit **23** and the first communication port **11** and between the slider unit **23** and the second communication port **12** are affected. Meanwhile, the first portion **231** and the second portion **232** are mounted on the guide frame **235**, and the connecting rod **22** can drive the first portion **231** and the second portion **232** to slide through the guide frame **235**, so that a maintenance cost can be reduced, and an utilization efficiency can be improved. When the guide frame **235** is abraded or the connecting rod **22** is damaged, only the lost part needs to be replaced, so that replacement and maintenance are facilitated, and the cost can be reduced.

(44) The pilot valve **110** of the reversible electromagnetic valve **100** is disposed on the valve body **10** and is respectively connected to the first capillary tube **141** and the second capillary tube **151**. And the pressure difference between the two sides of the piston unit **21** can be controlled by reversing the pilot valve **110**, thereby pushing the piston unit **21** to move.

(45) The working principle of the reversible electromagnetic valve **100** provided in the present disclosure is as follows: When the reversible electromagnetic valve **100** needs to be opened, the

high-pressure refrigerant is introduced into the first cavity **131** from the first capillary **141** through reversing of the pilot valve **110**, and the refrigerant in the second cavity **132** flows out of the second capillary **151**, so that the piston unit **21** is pushed to move towards the first communication port **11** and the second communication port **12**. And the piston unit **21** drives the slider unit **23** to move through the connecting rod **22** and the guide frame **235** to release the sealing/blocking of the first communication port **11** and the second communication port **12**.

(46) When the reversible electromagnetic valve **100** needs to be closed, the high-pressure refrigerant is introduced into the second cavity **132** from the second capillary **151** through reversing of the pilot valve **110**. The refrigerant in the first cavity **131** flows out of the first capillary **141**, so that the piston unit **21** is pushed to move towards the first communication port **11** and the second communication port **12**, and the piston unit **21** drives the slider unit **23** to move through the connecting rod **22** and the guide frame **235**, so that the first communication port **11** and the second communication port **12** are blocked/sealed.

(47) In the present disclosure, experiments verify that the single piston unit **21** is used for driving, the minimum action pressure difference is almost half of that of the two piston units **1013**. The reversible electromagnetic valve **100** in the present disclosure greatly improves the reliability of the action of the reversible electromagnetic valve **100** at a low pressure. This experiment was tested in the atmospheric environment.

(48) A friction force between the slider unit **23** and the valve base **16** is denoted as F_1 . A friction force between the sealing ring **312** and the valve core assembly **20** is denoted as F_2 . A friction force between the piston unit **21** and an inner wall of the valve body **10** is denoted as F_3 . An acting force between the medium in the medium cavity **133** and the valve core assembly **20** is denoted as F_4 . An acting force required by closing the reversible electromagnetic valve **100** is denoted as F_5 . And an acting force required for opening the reversible electromagnetic valve **100** is denoted as F_6 .

(49) Since the valve base **16** is provided with the first communication port **11** and the second communication port **12**, and the first communication port **11** and the second communication port **12** are both in contact with the air, the friction force F_1 between the slider unit **23** and the valve base **16** and the acting force F_4 between the medium in the medium cavity **133** and the valve core assembly **20** approach to 0. The acting force F_5 required by closing the reversible electromagnetic valve **100** is a product of a pressure of the medium in the second cavity **132** and a contact area between the piston unit **21** and the medium in the second cavity **132**, wherein the contact area between the piston unit **21** and the medium in the second cavity **132** is the area of the piston unit **21** facing a side surface of the second cavity **132**. The acting force F_6 required for opening the reversible electromagnetic valve **100** is a product of a pressure of the medium in the first cavity **131** and a contact area between the piston unit **21** and the medium in the first cavity **131**, wherein the contact area between the piston unit **21** and the medium in the first cavity **131** is a difference of the area of the piston unit **21** facing a side surface of the first cavity **131** minus a cross-sectional area of the connecting rod **22** perpendicular to an axis direction of the connecting rod **22**.

(50) Specifically, when the reversible electromagnetic valve **100** is closed, the friction force F_1 between the slider unit **23** and the valve base **16**, the friction force F_2 between the sealing ring **312** and the valve core assembly **20**, the friction force F_3 between the piston unit **21** and the inner wall of the valve body **10**, the acting force F_4 between the medium in the medium cavity **133** and the valve core assembly **20**, and the acting force F_5 required by closing the reversible electromagnetic valve **100** satisfy the following relationship: $F_5 > F_1 + F_2 + F_3 + F_4$. According to the above description, the friction force F_1 between the slider unit **23** and the valve base **16** and the acting force F_4 between the medium in the medium cavity **133** and the valve core assembly **20** approach to 0, the friction force F_2 between the sealing ring **312** and the valve core assembly **20**, the friction force F_3 between the piston unit **21** and the inner wall of the valve body **10**, and the acting force F_5 required by closing the reversible electromagnetic valve **100** satisfy the following relationship: $F_5 > F_2 + F_3$.

(51) When the reversible electromagnetic valve **100** is opened, the friction force F_1 between the slider unit **23** and the valve base **16**, the friction force F_2 between the sealing ring **312** and the valve core assembly **20**, the friction force F_3 between the piston unit **21** and the inner wall of the valve body **10**, the acting force F_4 between the medium in the medium cavity **133** and the valve core assembly **20**, the acting force F_5 required by closing the reversible electromagnetic valve **100**, and the acting force F_6 required for opening the reversible electromagnetic valve **100** satisfy the following relationship: $F_4 + F_6 > F_1 + F_2 + F_3 + F_5$. According to the above description, the friction force F_1 between the slider unit **23** and the valve base **16**, the acting force F_4 between the medium in the medium cavity **133** and the valve core assembly **20**, and the acting force F_5 required by closing the reversible electromagnetic valve **100** approach to 0, the friction force F_2 between the sealing ring **312** and the valve core assembly **20**, the friction force F_3 between the piston unit **21** and the inner wall of the valve body **10**, and the acting force F_6 required for opening the reversible electromagnetic valve **100** satisfy the following relationship: $F_6 > F_2 + F_3$.

(52) The valve opening pressure difference is mainly caused by resistance, and the resistance of the reversible electromagnetic valve **100** in the valve opening process is mainly the friction force F_2 between the sealing ring **312** and the valve core assembly **20** and the friction force F_3 between the piston unit **21** and the inner wall of the valve body **10**. The friction force F_2 between the sealing ring **312** and the valve core assembly **20** is less than the friction force F_3 between the piston unit **21** and the inner wall of the valve body **10**. It can be concluded from experiments that a minimum value of the valve opening pressure difference is approximately 0.11 MPa if the two piston units **1013** are used, however, a minimum value of the valve opening pressure difference is approximately 0.06 MPa in the reversible electromagnetic valve **100** in which the single piston unit **21** is used, and almost a half of that if the two piston units **1013** are used. Thus, compared to the traditional reversible electromagnetic valve with the two piston units, the reversible electromagnetic valve **100** in the present disclosure has less friction force and the valve opening pressure difference is less. Furthermore, a length of the valve core assembly **20** along the axial direction is less and a volume thereof is smaller, thereby decreasing the volume of space that needs to be occupied during installation, saving materials and reducing costs.

Embodiment 2

(53) Referring to FIG. 7, a structure, a working principle and a valve opening pressure difference verification principle of a reversible electromagnetic valve in the embodiment 2 is substantially the same as those of the reversible electromagnetic valve **100** in the embodiment 1, and so as, which are not repeated here, except that a structure of a valve body **10** in embodiment 2 is different.

(54) In this embodiment, the valve body **10** is separately disposed.

(55) The valve body **10** in this embodiment includes a valve base **16** and a sleeve pipe **40**. An intermediate end cover **30** is disposed between the valve base **16** and the sleeve pipe **40**. And the intermediate end cover **30** is fixedly connected to the valve body **10** and the sleeve pipe **40**, respectively.

(56) A first end cover **32** can be disposed at an end of the valve base **16** away from the sleeve pipe **40**. Alternatively, the first end cover **32** can be not additionally provided, and the valve base **16** and the first end cover **32** can be an integral structure. A connecting rod **22** penetrates through the intermediate end cover **30**. A slider unit **23** is located in the valve base **16** and can slide in the valve base **16**. The piston unit **21** is located in the sleeve pipe **40** and abuts against an inner wall of the sleeve pipe **40**, and the piston unit **21** is configured to drive the slider unit **23** to enable a first communication port **11** and a second communication port **12** to be in communication with each other or out of communication with each other under an action of a pressure difference between the two sides of the piston unit **21**.

Embodiment 3

(57) Referring to FIG. 8 and FIG. 9, the reversible electromagnetic valve includes a valve body **10**, a valve base **16**, a slider unit **23** and a connecting rod **22**. The valve base **16** is disposed in the valve

body **10**. The valve base **16** is provided with a valve port. The slider unit **23** is slidably disposed in the valve base **16** to close or open the valve port. And the connecting rod **22** and the slider unit **23** are in driving connection.

(58) The reversible electromagnetic valve includes a first sealing seat **50** and a second sealing seat **60**. The first sealing seat **50** is disposed in the valve body **10**. The first sealing seat **50** is provided with a first mounting groove **51**. An opening of the first mounting groove **51** can face away from the valve base **16** (or, away from the slider unit **23**). And the second sealing seat **60** is disposed in the first mounting groove **51**.

(59) The reversible electromagnetic valve **100** includes a first sealing ring **70** and a second sealing ring **80**. The first sealing ring **70** is disposed in the first sealing seat **50**. The second sealing ring **80** is disposed in the second sealing seat **60**. And the connecting rod **22** penetrates through the first sealing ring **70** and the second sealing ring **80**.

(60) By providing the first sealing seat **50** and the second sealing seat **60** and arranging the second sealing seat **60** in the first mounting groove **51**, the first sealing ring **70** is first mounted in the first sealing seat **50**, the second sealing ring **80** is then mounted in the second sealing seat **60**, and the second sealing seat **60** together with the second sealing ring **80** are mounted in the first mounting groove **51**. In this way, the first sealing ring **70** and the second sealing ring **80** can be respectively assembled into the first sealing seat **50** and the second sealing seat **60** more simply and quickly, so that the problem that the sealing ring is difficult to assemble into the sealing seat will be effectively solved.

(61) The intermediate end cover **30** of the reversible electromagnetic valve in embodiment 1 may include the first sealing seat **50** and the second sealing seat **60** in embodiment 3, and the following descriptions related to the first sealing seat **50** and the second sealing seat **60** may all be applied to embodiment 1.

(62) The first sealing seat **50** is further provided with a first through hole **52** and a first sealing groove **53**. The first through hole **52**, the first sealing groove **53** and the first mounting groove **51** are sequentially in communication with each other. Diameters of the first through hole **52**, the first sealing groove **53** and the first mounting groove **51** are sequentially increased. The first sealing ring **70** is arranged in the first sealing groove **53**. The second sealing seat **60** is provided with a second through hole **61**. An inner wall of the second through hole **61** is provided with a second sealing groove **611**. A diameter of the second through hole **61** is less than that of the first sealing groove **53**. The second sealing ring **80** is arranged in the second sealing groove **611**. The first through hole **52**, the first sealing groove **53** and the first mounting groove **51** are sequentially in communication with each other to facilitate the connecting rod **22** to pass through. The first sealing groove **53** is provided to facilitate accommodating and mounting of the first sealing ring **70**. By providing the second through hole **61** and a diameter of the second through hole **61** being smaller than a diameter of the first sealing groove **53**, it can be ensured that the connecting rod **22** smoothly passes, and the first sealing ring **70** can also be limited to prevent the first sealing ring **70** from entering the second through hole **61**. And the second sealing groove **611** is provided to facilitate accommodating and mounting of the second sealing ring **80**.

(63) Furthermore, a size of the first sealing groove **53** along an axial direction of the connecting rod **22** is greater than a size of the first sealing ring **70**. A diameter of the first through hole **52** is equal to that of the second through hole **61**. A diameter of the first sealing ring **70** is equal to that of the second sealing ring **80**. By providing the size of the first sealing groove **53** along an axial direction of the connecting rod **22** being greater than a size of the first sealing ring **70**, the first sealing ring **70** is capable of moving in the first sealing groove **53**. Because a pressure of one end of the first sealing seat **50** towards the valve base **16** is greater than that of the other end of the first sealing seat **50**, a thrust in a direction away from the valve base **16** can be generated, so that the first sealing ring **70** can be pushed to move towards the second sealing seat **60**. In this way, it can ensure that a sealing effect of the first sealing ring **70**, an area of the second sealing seat **60** being subject

to the thrust can be reduced, the second sealing seat **60** is prevented from being separated from the first sealing seat **50** by a large thrust, thereby improving a stability of the second sealing seat **60** during operation.

(64) Specifically, referring to FIG. **10**, a periphery of the opening of the first mounting groove **51** is provided with an annular flange **511**. The second sealing seat **60** is pressed into the first mounting groove **51** from an opening of the first mounting groove **51**. The annular flange **511** and the second sealing seat **60** are riveted. By providing the annular flange **511** being arranged on the periphery of the opening of the first mounting groove **51**, the first sealing seat **60** and the second sealing seat **60** can be riveted conveniently, and a connection therebetween is more stable and reliable by adopting a riveting connection mode. And the annular flange **511** can be a part of the first sealing seat **50**.

(65) Referring to FIG. **10** and FIG. **11**, before the annular flange **511** is riveted, a depth of the first mounting groove **51** is denoted as $L1$. An outer wall of the second sealing seat **60** is in a cylindrical shape. An axial length of the second sealing seat **60** is denoted as $L2$. The depth $L1$ of the first mounting groove **51** and the axial length $L2$ of the second sealing seat **60** satisfy the following relational expressions: $L1 > L2$. The annular flange **511** and an end surface of the second sealing seat **60** are riveted. By limiting the depth $L1$ of the first mounting groove **51** and the axial length $L2$ of the second sealing seat **60** in the above range, it can be ensured that the annular flange **511** and the end surface of the second sealing seat **60** are smoothly riveted.

(66) An inner wall of the first mounting groove **51** and an outer wall of the second sealing seat **60** can be in threaded connection, which is convenient to mount and dismount. Alternatively, the inner wall of the first mounting groove **51** and the outer wall of the second sealing seat **60** may also be laser welded. By adopting laser welding, the stability and reliability of the connection between the inner wall of the first mounting groove **51** and the outer wall of the second sealing seat **60** are improved.

(67) Furthermore, as shown in FIG. **12**, a third sealing groove **512** is formed in a bottom wall of the first mounting groove **51**. The third sealing groove **512** is arranged around the first sealing ring **70**. The reversible electromagnetic valve **100** further includes a third sealing ring **90**, and the third sealing ring **90** is arranged in the third sealing groove **512**. The third sealing ring **90** abuts against an end surface of the second sealing seat **60**. By providing the third sealing groove **512** in the bottom wall of the first mounting groove **51** and the third sealing ring **90** abutting against the end surface of the second sealing seat **60**, a sealing effect between the first sealing seat **50** and the second sealing seat **60** can be improved, and a thrust on the second sealing seat **60** due to high pressure can be reduced, thereby improving the stability of the second sealing seat **60** during operation. The sealing effect between the first sealing seat **50** and the second sealing seat **60** is prevented from being affected after the first sealing ring **70** fails.

(68) Furthermore, in the above embodiment, an outer wall of the first sealing seat **50** is provided with a welding groove **54**, and the outer wall of the first sealing seat **50** and the inner wall of the valve body **10** are welded. By providing the welding groove **54** on the outer wall of the first sealing seat **50**, it is convenient to weld the first sealing seat **50** with the inner wall of the valve body **10**.

(69) Referring to FIG. **13**, the present disclosure further provides an air conditioning unit **200**, including the reversible electromagnetic valve **100** described above.

(70) All the technical features of the above embodiments may be combined arbitrarily, so that the description is concise, and all possible combinations of the technical features in the above embodiments are not described, however, as long as there is no contradiction in the combination of these technical features, it should be considered that the scope of the present specification is set forth.

(71) The above embodiments only express several embodiments of the present disclosure, and the description thereof is more specific and detailed, but cannot be understood as a limitation on the application patent scope. It should be noted that, for a person of ordinary skill in the art, several variations and improvements may be made without departing from the concept of the present

disclosure, all of which fall within the protection scope of the present disclosure. Therefore, the scope of protection of the present disclosure shall be subject to the appended claims.

Claims

1. A reversible electromagnetic valve, comprising a valve body and a valve core assembly, wherein a first communication port and a second communication port are respectively formed in two sides of the valve body, the valve body is provided with a valve cavity, the valve core assembly is arranged in the valve cavity, and the valve core assembly is capable of slidably moving in the valve cavity to enable the first communication port and the second communication port to be in communication with each other or out of communication with each other; the valve core assembly comprises a piston unit, a connecting rod and a slider unit, the slider unit and the piston unit are respectively connected to two ends of the connecting rod, the slider unit is arranged adjacent to the first communication port and the second communication port, the first communication port and the second communication port are opposite to each other, the slider unit is configured to block the first communication port and the second communication port at the same time, and the piston unit is capable to be pushed to slide in the valve cavity due to a pressure difference between two sides of the piston unit; the reversible electromagnetic valve further comprises an intermediate end cover and a second end cover, wherein the intermediate end cover is fixed in the valve cavity, the piston unit is arranged on one side of the intermediate end cover away from the first communication port and the second communication port, a first cavity is defined by the intermediate end cover and the piston unit, and a second cavity is defined by the piston unit and the second end cover; the intermediate end cover comprises a first sealing seat and a second sealing seat, the first sealing seat is arranged in the valve body, the first sealing seat is provided with a first mounting groove, an opening of the first mounting groove is away from the slider unit, and the second sealing seat is arranged in the first mounting groove; the reversible electromagnetic valve further comprises a first sealing ring and a second sealing ring, the first sealing ring is arranged in the first sealing seat, the second sealing ring is arranged in the second sealing seat, and the connecting rod penetrates through the first sealing ring and the second sealing ring, and a diameter of the first sealing ring is equal to that of the second sealing ring.

2. The reversible electromagnetic valve of claim 1, further comprising a pilot valve, wherein the pilot valve is disposed on the valve body, the valve body is provided with a first hole and a second hole, the first hole is in communication with the first cavity, the second hole is in communication with the second cavity, the pilot valve is respectively in communication with the first hole and the second hole through capillary tubes, and the pilot valve is configured to control a pressure difference between the first cavity and the second cavity.

3. The reversible electromagnetic valve of claim 1, further comprising a first end cover, wherein the first end cover is disposed at an end of the valve body away from the second end cover, a medium cavity is defined between the first end cover and the intermediate end cover, the slider unit is disposed in the medium cavity, and when the reversible electromagnetic valve is in a first position, the first communication port and the second communication port are respectively in communication with the medium cavity.

4. The reversible electromagnetic valve of claim 1, wherein a side surface of the piston unit facing the intermediate end cover is provided with a first protrusion, and when the piston unit moves towards the first communication port and the second communication port, the first protrusion is capable of abutting against the intermediate end cover; and/or a side surface of the piston unit facing the second end cover is provided with a second protrusion, and when the piston unit moves towards the second end cover, the second protrusion is capable of abutting against the second end cover.

5. The reversible electromagnetic valve of claim 1, wherein a side surface of the piston unit facing

the intermediate end cover is provided with a first protrusion, the first protrusion is in an annular shape, a side surface of the first protrusion is provided with a first throttling hole for allowing a medium to flow out of a groove formed by the first protrusion; and/or a side surface of the piston unit facing the second end cover is provided with a second protrusion, the second protrusion is in an annular shape, a side surface of the second protrusion is provided with a second throttling hole for allowing a medium to flow out of a groove formed by the second protrusion.

6. The reversible electromagnetic valve of claim 1, wherein a side surface of the piston unit facing the intermediate end cover is provided with a first protrusion, the first protrusion is in an annular shape, a side surface of the first protrusion is provided with a first throttling groove for allowing a medium to flow out of a groove formed by the first protrusion; and/or a side surface of the piston unit facing the second end cover is provided with a second protrusion, the second protrusion is in an annular shape, a side surface of the second protrusion is provided with a second throttling groove for allowing a medium to flow out of a groove formed by the second protrusion.

7. The reversible electromagnetic valve of claim 1, wherein the slider unit comprises a first portion and a second portion opposite to each other, the first portion is capable of blocking the first communication port, the second portion is capable of blocking the second communication port, an accommodating cavity is formed between the first portion and the second portion, an elastic member is arranged in the accommodating cavity, and two ends of the elastic member abut against the first portion and the second portion respectively.

8. The reversible electromagnetic valve of claim 7, wherein the valve cavity comprises a medium cavity, the slider unit is arranged in the medium cavity, a balance hole is formed in the first portion and/or the second portion, and the medium cavity is in communication with the accommodating cavity through the balance hole; and/or the slider unit further comprises a guide frame, and the guide frame is sleeved on an outer side of the first portion and the second portion and connected to the connecting rod.

9. The reversible electromagnetic valve of claim 1, wherein the first sealing seat is further provided with a first through hole and a first sealing groove, the first through hole, the first sealing groove and the first mounting groove are sequentially in communication with each other, diameters of the first through hole, the first sealing groove and the first mounting groove are sequentially increased, and the first sealing ring is arranged in the first sealing groove; the second sealing seat is provided with a second through hole, an inner wall of the second through hole is provided with a second sealing groove, a diameter of the second through hole is less than that of the first sealing groove, and the second sealing ring is arranged in the second sealing groove.

10. The reversible electromagnetic valve of claim 9, wherein a size of the first sealing groove along an axial direction of the connecting rod is greater than a size of the first sealing ring, a diameter of the first through hole is equal to that of the second through hole.

11. The reversible electromagnetic valve of claim 1, wherein an inner wall of the first mounting groove is in threaded connection with an outer wall of the second sealing seat; and/or an outer wall of the first sealing seat is provided with a welding groove.

12. An air conditioning unit, comprising the reversible electromagnetic valve of claim 1.

13. A reversible electromagnetic valve, comprising a valve body and a valve core assembly, wherein a first communication port and a second communication port are respectively formed in two sides of the valve body, the valve body is provided with a valve cavity, the valve core assembly is arranged in the valve cavity, and the valve core assembly is capable of slidably moving in the valve cavity to enable the first communication port and the second communication port to be in communication with each other or out of communication with each other; the valve core assembly comprises a piston unit, a connecting rod and a slider unit, the slider unit and the piston unit are respectively connected to two ends of the connecting rod, the slider unit is arranged adjacent to the first communication port and the second communication port, the first communication port and the second communication port are opposite to each other, the slider unit is configured to block the first

communication port and the second communication port at the same time, and the piston unit is capable to be pushed to slide in the valve cavity due to a pressure difference between two sides of the piston unit; the reversible electromagnetic valve further comprises an intermediate end cover and a second end cover, wherein the intermediate end cover is fixed in the valve cavity, the piston unit is arranged on one side of the intermediate end cover away from the first communication port and the second communication port, a first cavity is defined by the intermediate end cover and the piston unit, and a second cavity is defined by the piston unit and the second end cover; the intermediate end cover comprises a first sealing seat and a second sealing seat, the first sealing seat is arranged in the valve body, the first sealing seat is provided with a first mounting groove, an opening of the first mounting groove is away from the slider unit, and the second sealing seat is arranged in the first mounting groove; the reversible electromagnetic valve further comprises a first sealing ring and a second sealing ring, the first sealing ring is arranged in the first sealing seat, the second sealing ring is arranged in the second sealing seat, and the connecting rod penetrates through the first sealing ring and the second sealing ring; a third sealing groove is formed in a bottom wall of the first mounting groove, the third sealing groove is arranged around the first sealing ring, the reversible electromagnetic valve further comprises a third sealing ring, the third sealing ring is arranged in the third sealing groove, and the third sealing ring abuts against an end surface of the second sealing seat.

14. A reversible electromagnetic valve, comprising: a valve body, a valve seat, a slider unit and a connecting rod, wherein the valve seat is disposed in the valve body, the valve seat is provided with a valve port, and the slider unit is slidably disposed in the valve seat to close or open the valve port, the connecting rod is in driving connection with the slider unit; a first sealing seat and a second sealing seat, wherein the first sealing seat is arranged in the valve body, the first sealing seat is provided with a first mounting groove, an opening of the first mounting groove is away from the valve seat, and the second sealing seat is arranged in the first mounting groove; a first sealing ring and a second sealing ring, wherein the first sealing ring is arranged in the first sealing seat, the second sealing ring is arranged in the second sealing seat, and the connecting rod penetrates through the first sealing ring and the second sealing ring, and a diameter of the first sealing ring is equal to that of the second sealing ring.

15. The reversible electromagnetic valve of claim 14, wherein the first sealing seat is further provided with a first through hole and a first sealing groove, the first through hole, the first sealing groove and the first mounting groove are sequentially in communication with each other, diameters of the first through hole, the first sealing groove and the first mounting groove are sequentially increased, and the first sealing ring is arranged in the first sealing groove; the second sealing seat is provided with a second through hole, an inner wall of the second through hole is provided with a second sealing groove, a diameter of the second through hole is less than that of the first sealing groove, and the second sealing ring is arranged in the second sealing groove.

16. The reversible electromagnetic valve of claim 15, wherein a size of the first sealing groove along an axial direction of the connecting rod is greater than a size of the first sealing ring, a diameter of the first through hole is equal to that of the second through hole.

17. The reversible electromagnetic valve of claim 14, wherein an inner wall of the first mounting groove is in threaded connection with an outer wall of the second sealing seat; and/or an outer wall of the first sealing seat is provided with a welding groove.

18. The reversible electromagnetic valve of claim 14, wherein a third sealing groove is formed in a bottom wall of the first mounting groove, the third sealing groove is arranged around the first sealing ring, the reversible electromagnetic valve further comprises a third sealing ring, the third sealing ring is arranged in the third sealing groove, and the third sealing ring abuts against an end surface of the second sealing seat.
